AIR POLLUTION ANALYSIS USING K-MEANS CLUSTERING AND PCA

A Project Report

Submitted by

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In partial fulfilment for the requirements of the project

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CERTIFICATE

This is to certify that the project work entitled **"Air Pollution Analysis using K-Means Clustering and PCA"** is a Bonafide record of project work carried out by the following students:

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• from the **Department of Computer Science and Engineering, SRM AP University** .The students conducted this project work under my supervision during the period **January 2026 to june 2026.**

It is further certified that, to the best of my knowledge, this project has not previously formed the basis for the award of any degree or any similar title to this or any other candidate.

This is also to certify that the project work represents the **teamwork of the candidates** and demonstrates their ability to apply **machine learning techniques such as K-Means clustering and Principal Component Analysis (PCA)** for analyzing real-world air pollution data.

The project includes data preprocessing, feature selection, clustering, visualization, and interpretation of pollution levels into meaningful categories such as **Low, Moderate, and High Pollution**, which reflects the practical understanding and implementation skills of the students.

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1. INTRODUCTION

In recent years, air pollution has emerged as one of the most serious environmental challenges affecting human health, climate, and overall quality of life. Rapid industrialization, urbanization, population growth, and increased vehicle usage have significantly contributed to the rise in air pollution levels across cities. Pollutants such as particulate matter (PM2.5, PM10), nitrogen dioxide (NO2), sulfur dioxide (SO2), and carbon monoxide (CO) are commonly found in the atmosphere and pose severe risks to respiratory and cardiovascular health.

Monitoring and analyzing air quality data is essential for understanding pollution trends and implementing effective control measures. However, the large volume of data generated from air quality monitoring stations makes manual analysis difficult, time-consuming, and inefficient. The data is often complex, containing multiple features, missing values, and varying scales, which makes it challenging to interpret meaningful patterns directly.

With the advancement of data science and machine learning techniques, it has become possible to analyze large datasets efficiently and extract valuable insights. Machine learning provides automated methods for identifying hidden patterns, grouping similar data points, and making data-driven decisions. Among these techniques, clustering plays a vital role in unsupervised learning, where data is grouped based on similarity without predefined labels.

This project focuses on applying the K-Means clustering algorithm to analyze air pollution data. K-Means is a widely used clustering technique that partitions data into distinct groups based on feature similarity. By grouping similar pollution levels together, it becomes easier to understand the distribution and intensity of air pollution across different conditions.

In addition to clustering, this project also utilizes Principal Component Analysis (PCA), a dimensionality reduction technique that transforms high-dimensional data into a lower-dimensional space. PCA helps in visualizing complex datasets by reducing them to two principal components, making it easier to interpret cluster formations and relationships between pollutants.

The dataset used in this project contains various air quality parameters such as PM2.5, PM10, NO2, SO2, and CO. These features are selected as they are key indicators of air pollution. The project involves multiple steps including data preprocessing, feature selection, normalization, clustering, and visualization.

By applying K-Means clustering, the dataset is divided into different groups representing varying pollution levels. These clusters are further interpreted and labeled as Low, Moderate, and High Pollution categories. This classification helps in understanding the severity of pollution and provides a clearer picture of environmental conditions.

Furthermore, the use of visualization techniques such as histograms, correlation heatmaps, and PCA scatter plots enhances the interpretability of the results. These visual tools help in identifying relationships between pollutants and understanding how different variables contribute to pollution levels.

Overall, this project demonstrates how machine learning techniques can be effectively used to analyze real-world environmental data. It highlights the importance of data preprocessing, feature engineering, and model selection in achieving meaningful insights. The results obtained from this analysis can be useful for environmental monitoring, policy-making, and raising awareness about air pollution.

2. PROBLEM DEFINITION

In today's world, air pollution data is collected continuously from various monitoring stations. This data contains multiple pollutant parameters recorded over time. Although this data is highly valuable, it remains underutilized due to its complexity and volume.

The major challenges associated with air pollution data include:

- The dataset contains **missing and inconsistent values**, which affects analysis accuracy.
- The presence of multiple pollutants makes it difficult to identify dominant factors manually.
- Data is **high-dimensional**, making visualization and interpretation complex.
- Pollution levels vary over time and location, leading to **unclear patterns**.

- Traditional methods fail to group pollution data into meaningful categories.

Due to these challenges, there is a need for an intelligent system that can process and analyze pollution data automatically. Machine learning techniques such as clustering can help in grouping similar pollution conditions and identifying patterns effectively.

3. PROBLEM STATEMENT

The aim of this project is to design and implement a machine learning-based system that analyzes air pollution data and groups it into meaningful clusters.

The system should:

- Accept air pollution dataset as input
- Perform data cleaning and preprocessing
- Select important pollutant features
- Normalize the data to ensure consistency
- Apply K-Means clustering algorithm
- Determine optimal number of clusters using Elbow Method
- Visualize clusters using PCA
- Categorize pollution levels into Low, Moderate, and High

The system should be efficient, scalable, and capable of handling real-world environmental data.

4. OBJECTIVES

4.1 Technical Objectives

- To understand data preprocessing techniques such as handling missing values and normalization
- To select relevant pollutant features from the dataset
- To implement K-Means clustering algorithm from scratch

- To apply PCA for dimensionality reduction
- To use visualization tools like histograms and heatmaps
- To evaluate clustering results using inertia (Elbow Method)

4.2 Functional Objectives

- To group air pollution data into clusters
- To analyze pollutant distributions
- To identify relationships between pollutants
- To classify pollution levels into categories

4.3 Academic Objectives

- To gain hands-on experience in machine learning
- To understand unsupervised learning techniques
- To apply theoretical concepts to real-world data
- To build a structured end-to-end data analysis pipeline

5. METHODOLOGY

The methodology of this project follows a structured data science workflow:

5.1 Data Collection

The dataset used in this project contains air pollution measurements such as PM2.5, PM10, NO2, SO2, and CO. These pollutants are key indicators of air quality.

The dataset consists of approximately **59,000+ records**, making it suitable for analysis using machine learning techniques.

5.2 Data Preprocessing

Raw data often contains noise and missing values. Therefore, preprocessing is necessary.

Steps performed:

- Removed missing values using dropna()
- Converted data into numeric format
- Filled missing values using mean
- Selected important features (PM2.5, PM10, NO2, SO2, CO)
- Cleaned and structured dataset

This step ensures that the data is ready for further analysis.

5.3 Feature Selection

Not all columns are useful for clustering. Therefore, only relevant pollutant features were selected:

- PM2.5
- PM10
- NO2
- SO2
- CO

These features represent the major contributors to air pollution.

5.4 Data Normalization

Since different pollutants have different units and scales, normalization is required.

- Standardization was applied using mean and standard deviation
- This ensures all features contribute equally

5.5 Data Visualization

Visualization helps in understanding data distribution:

- Histograms were used to study pollutant distribution
- Heatmap was used to identify correlation between pollutants

Observations:

- PM2.5 and PM10 show strong correlation
- Other pollutants show moderate relationships

5.6 K-Means Clustering

K-Means clustering algorithm was implemented from scratch.

Steps:

1. Initialize random centroids
2. Assign each data point to nearest centroid
3. Update centroids
4. Repeat until convergence

This groups data into clusters based on similarity.

5.7 Elbow Method

To find optimal number of clusters:

- Calculated inertia for different values of K
- Plotted graph (K vs Inertia)
- Identified elbow point

Optimal clusters selected: **K = 3**

5.8 PCA (Principal Component Analysis)

PCA was used to reduce dimensionality:

- Converted 5D data into 2D
- Helped visualize clusters clearly
- Improved understanding of data structure

5.9 Cluster Labeling

Clusters were labeled based on pollution intensity:

- Cluster 0 → Low Pollution
- Cluster 1 → Moderate Pollution
- Cluster 2 → High Pollution

6. RESULTS AND ANALYSIS

The results of the project demonstrate effective clustering of air pollution data.

6.1 Cluster Formation

- Data was divided into 3 clusters
- Each cluster represents different pollution levels

6.2 Visualization Results

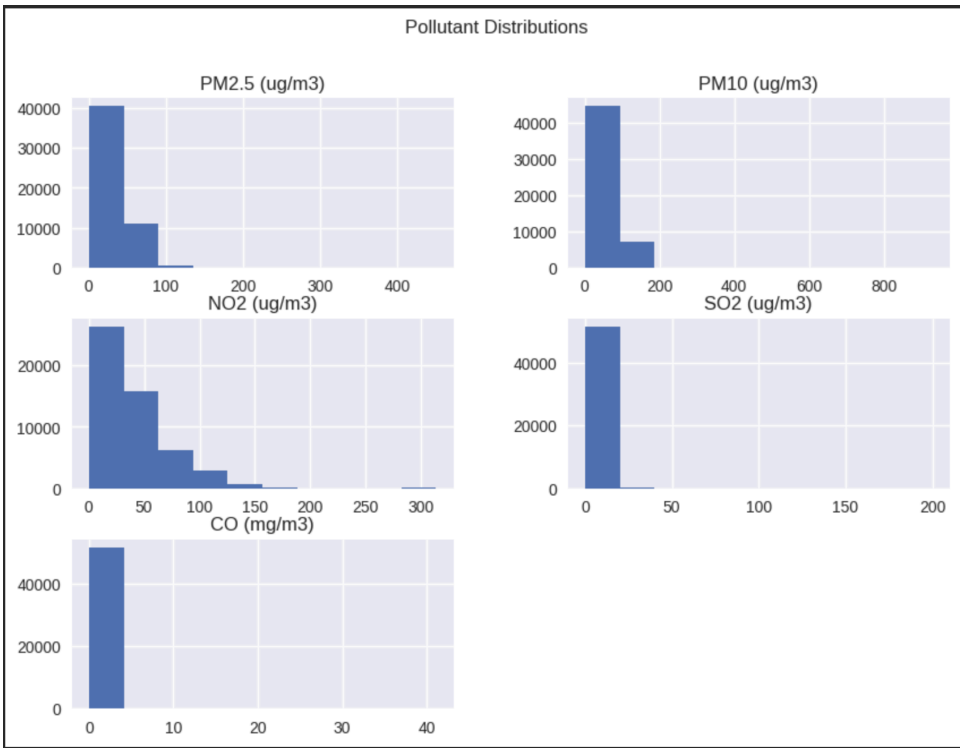
- Histogram shows distribution of pollutants
- Heatmap shows correlation
- PCA plot shows clear cluster separation

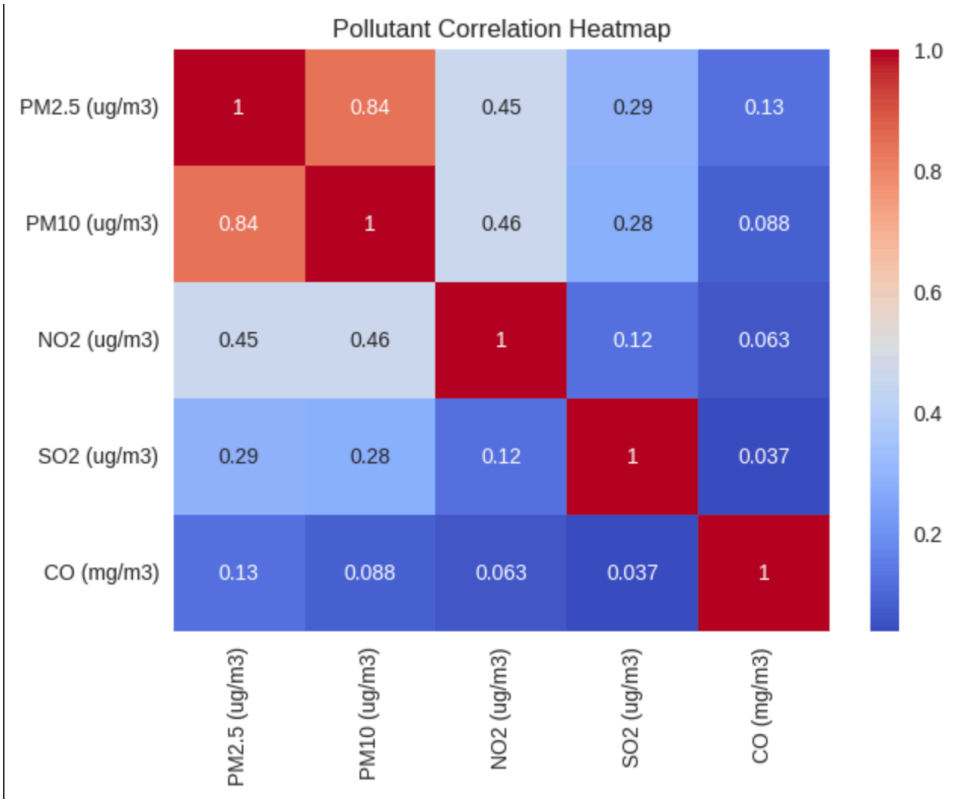
6.3 Observations

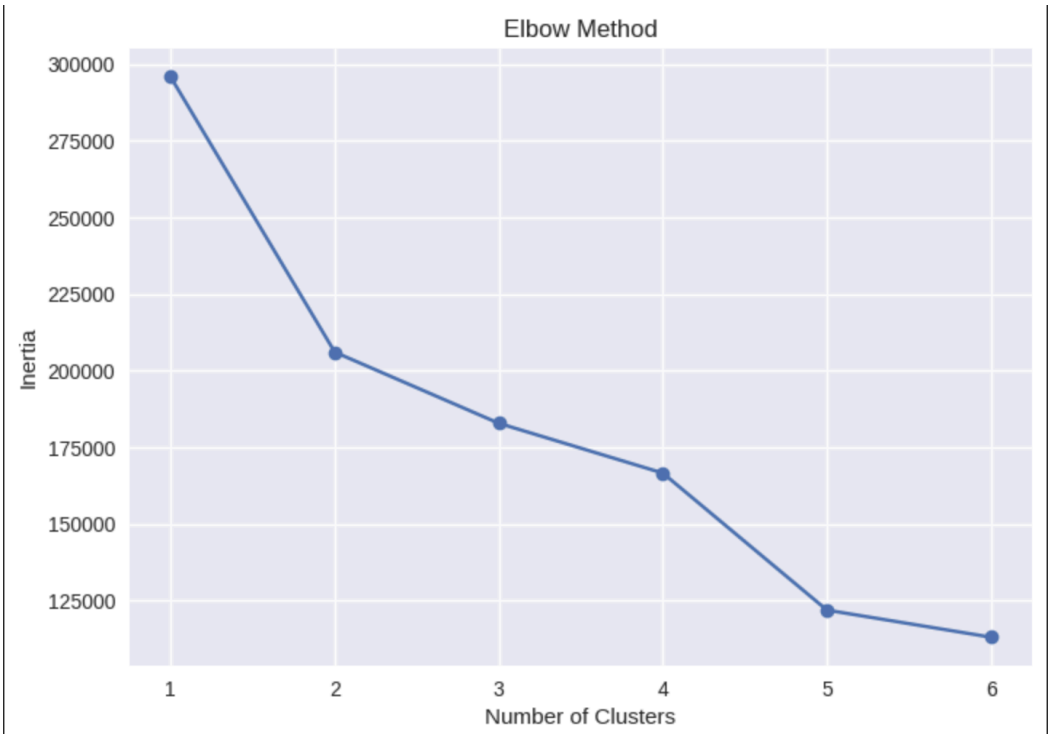
- PM2.5 and PM10 are dominant pollutants
- High pollution cluster has wider spread
- Moderate cluster lies between low and high

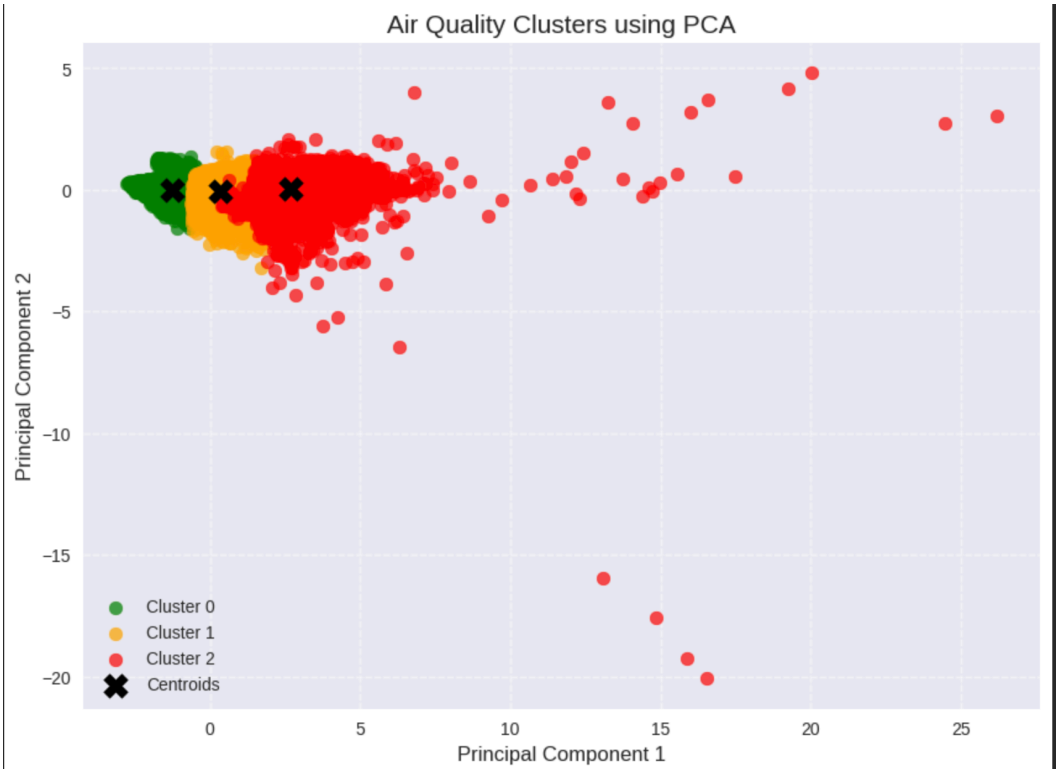
6.4 Cluster Interpretation

- Cluster 0 → Clean air conditions
- Cluster 1 → Moderate pollution levels
- Cluster 2 → High pollution (dangerous levels)


Pollutant Distributions
PM2.5 (ug/m3)
PM10 (ug/m3)
NO2 (ug/m3)
SO2 (ug/m3)
CO (mg/m3)


Pollutant Correlation Heatmap
PM2.5 (ug/m3)
PM10 (ug/m3)
NO2 (ug/m3)
SO2 (ug/m3)
CO (mg/m3)
1 0.84 0.45 0.29 0.13
0.84 1 0.46 0.28 0.088
0.45 0.46 1 0.12 0.063
0.29 0.28 0.12 1 0.037
0.13 0.088 0.063 0.037 1
1.0
0.8
0.6
0.4
0.2


Elbow Method
Inertia
300000
275000
250000
225000
200000
175000
150000
125000
1
2
3
4
5
6
Number of Clusters


Air Quality Clusters using PCA
Principal Component 2
5
0
−5
−10
−15
−20
Cluster 0
Cluster 1
Cluster 2
Centroids
0
5
10
15
20
25
Principal Component 1

7. FUTURE WORK

This project can be extended in multiple ways:

- Use advanced clustering algorithms like DBSCAN
- Apply Deep Learning models
- Develop real-time air quality monitoring system
- Include more features such as weather data
- Build web-based dashboard for visualization
- Predict future pollution levels

8. CONCLUSION

This project successfully demonstrates the use of machine learning techniques to analyze air pollution data.

K-Means clustering effectively groups pollution levels into meaningful categories. PCA helps visualize clusters and improves understanding of patterns.

The project highlights the importance of preprocessing, feature selection, and visualization in data analysis. It also shows how unsupervised learning can be applied to real-world problems.

Overall, the system provides valuable insights into pollution patterns and can be useful for environmental monitoring and decision-making.

9. REFERENCES

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